

Ulverton Wool Mill Safety Manual

1 Introduction

The Ulverton Wool Mill is a historic company which has industrial artifacts from the last 150 years. Their highlighted attraction is a 17th-century Davis and Furber wool spinning machine. This machine is capable of taking larger spools of wool and twining them into yarn. Because of the age of the device, safety was not a major motivator in the design. There are many scenarios in which an operator or technician could harm themselves if not careful. This document includes safety protocols for the general operation of the machine, along with safety guidelines for technicians to follow while preparing the machine.

2 Workplace Safety

Textile mills are often dangerous if proper precautions are not taken. Due to the many moving parts of most machines in the yarn-making industry, anything could go wrong at any moment. Because of this, it is essential to understand the different injuries that can occur so they can be prevented. Within a textile mill, there are four main ways injury can occur. The first is with contact with machinery. This is the most dangerous due to the moving parts. Because of the moving parts, components such as belts, gears, and wires, general contact will cause some injury. Exposure to fibre and dust is also a substantial concern within a textile mill. Due to the different materials and processes, airborne toxins can cause long-term harm if masks are not worn when appropriate. Textile mills are also very loud when their machines are in use. Wearing ear protection can help prevent hearing loss and tinnitus. The last safety concern within a textile mill is overexertion. Some applications in the yarn industry can require heavy lifting and other demanding actions. Completing these actions repeatedly over time can lead to muscle and joint issues. Below in Table 1 is a complete list of each of the main concerns with their respective list of possible injuries.

Table 1: Textile Mill Injuries¹

Areas of Concern	Possible Injuries
Contact with machinery	Serious Crushing Tearing Cuts Burns Hair getting stuck Amputation of fingers, hands, or arms
Exposure to fibres and dust	Respiratory Issues Prolonged exposure can lead to cancer
Exposure to noise	Hearing Damage Tinnitus
Overexertion	Muscle and joint stress/pain Sprains Tearing Trauma

A textile mill is similar to most businesses since it has employees and a workforce. Because of this, it is also important to focus on the general workplace protocols for safety. An agency that enforces workplace safety is the Occupational Safety and Health Administration (OSHA)². OSHA is responsible for ensuring that employers uphold safety for their employees. This includes physical but also social safety. Within the Ulverton Wool Mill, OSHA's guidelines must be followed to provide its employees with a safe environment. Below in Table 2 is a list of the requirements, how to meet the requirements, and how to enhance them.

Table 2: OSHA Requirements²

OSHA Requirements	Workplace safety programs Recordkeeping and reporting A safe Workplace PPE Report Hazards Training Filing a complaint Participating in OSHA inspections
Meeting the Requirements	Have regular safety audits Be up to date with OSHA's standards Engage employees
How to Enhance Requirements	Make recording and reporting easy for all employees Strengthen communication and training Empower employees Make support easy to obtain

3 Technician Safety

The most crucial safety to abide by is safety for technicians. Ulverton Wool Mill often has technicians come in to repair the wool spinning machine. Even when the machine is turned off it can offer hazards due to its many mechanical pieces. Because of this, any technicians who come to repair the machine must follow all respective guidelines. This also applies to anyone else who is inspecting, repairing, or tampering with the machine. There are three areas of safety that technicians must be aware of. The first is personal protective equipment (PPE). This includes all protective equipment that is required for the task at hand. In the case that a technician does not know what PPE to wear, they should ask a supervisor or wear as much as possible. The next area of safety is safe work procedures. These guidelines are based on developing safety habits in the workplace that prevent injury. Emergency preparation is the last area of safety for technicians. When a technician goes to an area they are unfamiliar with, they must gain an understanding of what to do in an emergency. The main guideline for emergency preparation is to understand where all fire exits and extinguishers. Below in Table 3 are each of the areas of safety with their respective safety guidelines.

Table 3: Safety Guidelines for Technicians³

Area of Safety	Safety Guidelines
PPE (If Applicable)	• Hard hat • Safety glasses • Steel-toed boots • Safety gloves • Hearing protection • Respiratory protection
Safe Work Procedures	• Check tools and equipment routinely for damage • Use the most appropriate tools for the job • Bend the knees and use legs when lifting • Effective communication • Notify supervisor immediately if there are possible dangers
Emergency Preparation	• Be aware of fire exits and extinguishers • Have easy access to emergency contacts • Try to have someone with first-aid certification

An example of danger with the wool spinning machine is the headstock. The headstock is the component that takes the input power and distributes it between the machine's components. This part of the machine has the most moving parts and can cause harm if not careful. The first area of concern is the moving gears. When in motion, these gears can pull in shirts, hair, and even fingers. Having a finger get pulled into a gear mechanism will likely cause permanent damage or loss of the finger. Another area of concern is the tension ropes. Many of these ropes move when the machine is operating, and touching one will likely cause a friction burn. Lastly, some components are heavy, and when removed from the machine, can be a source of danger. Below in Figure 1 is an annotated picture of the headstock and its many dangers.

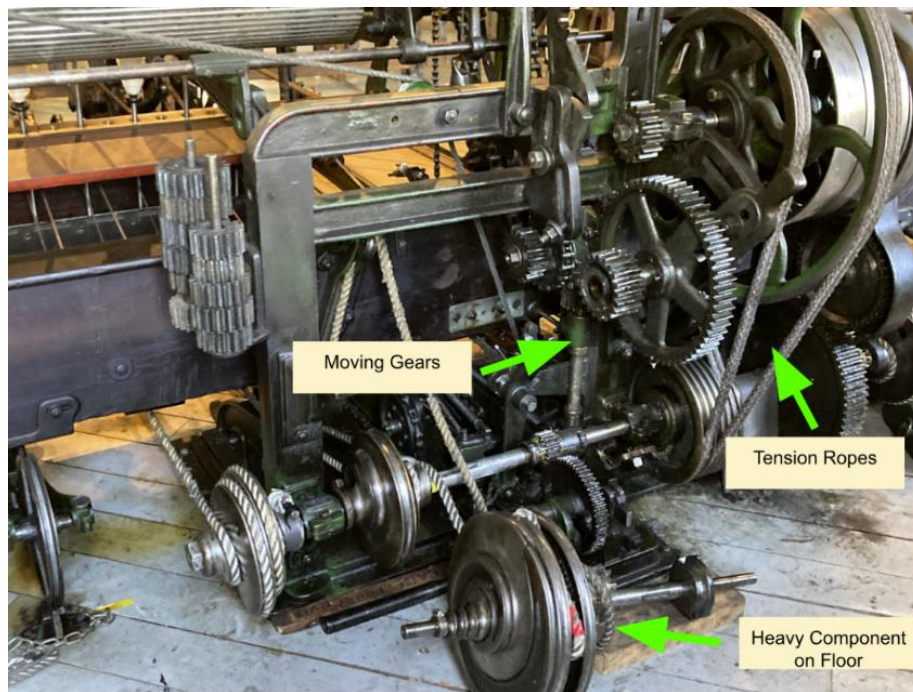


Figure 1: Ulverton Headstock Dangers

Works Cited

- [1] “Common Injuries for Textile Mill Employees.” Hardison & Cochran. <https://www.lawyernc.com/workers-compensation-lawyer/common-workplace-injuries/textile-mill-employees/> [Accessed November 20 2025].
- [2] “OSHA Compliance Requirements: A Detailed Overview.” SiteDocs. <https://www.sitedocs.com/blog/osha-compliance-requirements/> [Accessed November 20 2025].
- [3] “Effective Safety Tips For Construction Technicians.” Philadelphia Technician Training Institute. <https://ptt.edu/effective-safety-tips-for-construction-technicians/> [Accessed November 21 2025].